
Cornerstones or Stumbling Blocks? Deciphering the Rock Tokens in On-Policy Distillation

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Abstract

1 While recent work in Reinforcement Learning with Verifiable Rewards (RLVR)
2 has shown that a small subset of critical tokens disproportionately drives reasoning
3 gains, an analogous token-level understanding of On-Policy Distillation (OPD)
4 remains largely unexplored. In this work, we investigate high-loss tokens, a token
5 type that—as the most direct signal of student-teacher mismatch under OPD’s per-
6 token KL objective—should progressively diminish as training converges according
7 to existing studies; however, our empirical analysis shows otherwise. Even after
8 OPD training reaches apparent saturation, a substantial subset of tokens continues
9 to exhibit persistently high loss; these tokens, which we term Rock Tokens, can
10 account for up to 50% of the tokens in generated outputs. Our investigation reveals
11 two startling paradoxes. First, despite their high occurrence frequency providing
12 a disproportionately large share of total gradient norms, Rock Tokens themselves
13 remain stagnant throughout training, resisting teacher-driven corrections. Second,
14 through causal intervention, we find that these tokens provide negligible func-
15 tional contribution to the model’s actual reasoning performance. These findings
16 suggest that a vast amount of optimization bandwidth is spent on structural and
17 discourse residuals that the student model cannot or need not internalize. By de-
18 constructing these dynamics, we demonstrate that strategically bypassing these
19 “stumbling blocks” can significantly streamline the alignment process, challenging
20 the necessity of uniform token weighting and offering a more efficient paradigm
21 for large-scale model distillation.

22 1 Introduction

23 On-Policy Distillation (OPD) has established itself as a cornerstone of the modern post-training
24 pipeline for Large Language Models (LLMs). By aligning a student model with a superior teacher
25 through trajectories sampled from its own policy, OPD enables a token level of reasoning refinement
26 that goes beyond static Supervised Fine-Tuning (SFT). This effectiveness has been validated by
27 industrial works such as DeepSeek-V4 [2], MiMo [18], and Qwen-3 [20], where OPD serves as
28 a vital component alongside SFT or Reinforcement Learning with Verifiable Rewards (RLVR) to
29 further squeeze out reasoning performance.

30 Recent studies in Reinforcement Learning with Verifiable Rewards (RLVR) have revealed that not
31 all tokens contribute equally to model learning: a small subset of critical tokens, such as high-
32 entropy “forking tokens” at decisive branching points, disproportionately drives reasoning gains [16].
33 While this token-level perspective has reshaped our understanding of RLVR dynamics, an analogous
34 investigation in OPD remains largely unexplored, despite OPD’s reliance on dense token-level
35 supervision. In this work, we initiate such an investigation by focusing on a token type uniquely
36 meaningful to the OPD paradigm: high-loss tokens. Unlike RLVR, where token importance is
37 inferred from policy entropy or reward signals, OPD provides a direct measure of student-teacher

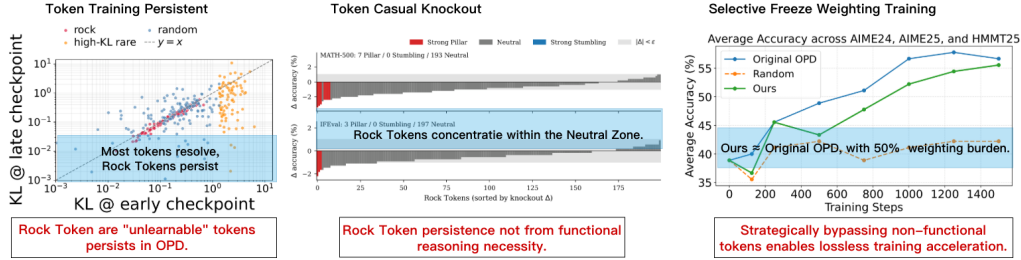


Figure 1: **The lifecycle and functional impact of Rock Tokens in OPD.** (a) **Phenomenon:** Identification of optimization-resistant tokens. (b) **Mechanism:** Causal evidence of structural redundancy via token knock-out. (c) **Utility:** Performance parity achieved through strategic gradient sparsification.

mismatch through per-token KL divergence; high-loss tokens thus explicitly mark the positions where the student’s policy diverges most sharply from the teacher’s, making them natural candidates for the “critical correction points” that drive alignment [12].

Intuitively, as the student aligns with the teacher during the OPD process, both the magnitude and cardinality of these high-loss tokens are expected to diminish, eventually leading to an optimization plateau. However, our empirical investigations reveal a striking phenomenon: even after training has reached apparent saturation, a stubborn population of tokens—which we formally define as Rock Tokens—continues to exhibit consistently high loss. Notably, while these tokens represent a small minority of the vocabulary (approximately 6%), they account for as much as 50% of the total token frequency in the model’s output. This persistence challenges conventional understanding of model convergence and raises a fundamental question: Are Rock Tokens indispensable Pillars for policy alignment, or are they Stumbling Blocks that create unnecessary optimization redundancy?

To provide a systematic understanding of the Rock Token phenomenon, our research is structured across three investigative dimensions: identifying **what** are rock tokens, analyzing **why** they emerge and persist, and uncovering **how** they collectively impact the overall OPD training process. This comprehensive framework is operationalized through the following three stages:

First, we characterize the identity of Rock Tokens (What). We begin by identifying Rock Tokens linguistic and statistical properties, revealing that they primarily comprise **syntactic and structural scaffolding**—such as formatting delimiters, whitespace symbols, and high-frequency discourse markers (e.g., “So”, “Wait”). To determine the dynamic role of these elements, we employ quantitative gradient dynamics analysis to address **RQ1: Do Rock Tokens still provide useful learning signals during optimization?** By examining whether their persistent high loss effectively translates into constructive gradients, we assess whether these tokens offer steady directional guidance or merely represent stagnant optimization noise throughout the OPD process.

Second, we investigate the underlying mechanisms behind their persistence (Why). Beyond the actual contribution of Rock Tokens during training is clarified, a more profound question emerges: **why do these tokens persist in the first place?** We hypothesize that the student model develops a strong decoding dependency on these specific tokens to maintain its reasoning flow, creating a path dependency that resists corrections from the teacher model. This leads to our **RQ2: Does Rock Token persistence stem from student path dependency?** To investigate this, we conduct *token knock-out* experiments as a form of ablation analysis, examining whether the student’s logical coherence is so deeply intertwined with these persistent residuals that it becomes structurally unable to shift its policy toward the teacher’s distribution.

Finally, we evaluate how Rock Tokens influence the overall OPD training process (How). Finally, we arrive at a provocative question: what would happen if these tokens were excluded from the very beginning of OPD training? This leads to our final investigation, **RQ3: What is the genuine functional contribution of Rock Tokens to model training?** By freezing the gradients of these tokens from the onset of training, we intuitively isolate their functional necessity throughout the entire distillation process. This causal intervention allows us to rigorously determine whether Rock Tokens serve as indispensable pillars of the model’s reasoning integrity or if they represent a strategic opportunity to bypass computational redundancy and optimize the efficiency of the alignment pipeline.

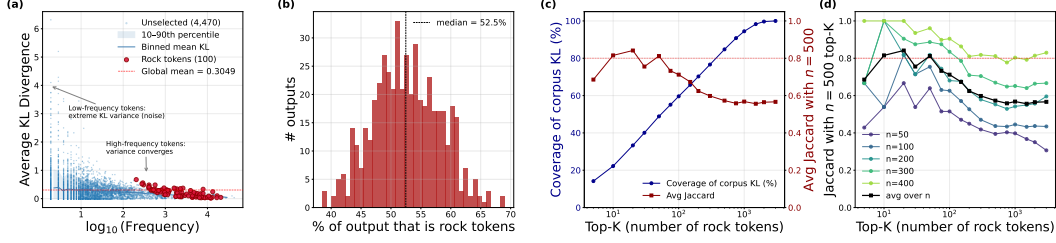


Figure 2: **Empirical identification and stability of Rock Tokens.** (a) Per-token KL $\hat{\ell}_v$ vs. frequency on $N=500$ MATH-500 trajectories: rare tokens are noise-dominated, while the Rock Score $R(v)$ isolates true Rock Tokens (red) at the upper edge of stable frequency bands. (b) Per-sequence Rock-Token density (median 52.5%). (c) Cumulative loss coverage (blue) and selection stability (red) vs. cutoff K ; $K=100$ balances reproducibility with $\approx 60\%$ coverage of the distillation burden. (d) Jaccard ranking robustness across sample sizes $n \in [50, 400]$: the top-100 selection is stable, larger cutoffs degrade.

79 In summary, our key contributions are as follows:

- 80 • For the first time, we identify and characterize **Rock Tokens**, a persistent high-loss phenomenon
- 81 that unveils a previously hidden aspect of optimization dynamics in On-Policy Distillation.
- 82 • We deconstruct the intricate dynamics of the OPD process by systematically uncovering the
- 83 functional roles of Rock Tokens across both the training and inference phases.
- 84 • Our research reveals a critical inefficiency in current alignment pipelines, demonstrating that a sub-
- 85 stantial portion of training signals in OPD provides significantly lower-than-expected contributions
- 86 to the model’s ultimate reasoning performance.

87 2 Rock Tokens

88 2.1 Token-Level Loss in On-Policy Distillation

89 On-policy distillation (OPD) optimizes a student policy π_θ by matching a teacher policy π_T on
 90 trajectories sampled from the student’s own distribution. The objective minimizes the per-token
 91 reverse KL divergence:

$$\mathcal{L}_{\text{OPD}}(\theta) = \mathbb{E}_{x_{1:T} \sim \pi_\theta} \left[\sum_{t=1}^T D_{\text{KL}}(\pi_\theta(\cdot | x_{<t}) \| \pi_T(\cdot | x_{<t})) \right]. \quad (1)$$

92 The loss at position t is $\ell_t = \log \pi_\theta(x_t | x_{<t}) - \log \pi_T(x_t | x_{<t})$, measuring the student-teacher
 93 mismatch [4]. Ideally, as the student aligns with the teacher, the magnitude of ℓ_t should systematically
 94 diminish across the vocabulary, leading to an optimization plateau where corrective signals eventually
 95 vanish.

96 2.2 Defining and Detecting Rock Tokens

97 Contrary to the expected convergence, we observe a subset of tokens that persistently exhibit high
 98 KL divergence even after training saturation. We term these **Rock Tokens**.

99 At an intuitive level, our detection method isolates these tokens based on their *Total Distillation*
 100 *Burden*—the absolute volume of optimization capacity consumed by a specific vocabulary item across
 101 the entire training corpus. Since the loss $\mathcal{L}_{\text{OPD}}$ is an expectation over student trajectories, it admits
 102 an exact additive decomposition over token types v in the vocabulary $\mathcal{V}$:

$$\mathcal{L}_{\text{OPD}}(\theta) = \mathbb{E}_{x \sim \pi_\theta} \left[\sum_t \ell_t \right] = \sum_{v \in \mathcal{V}} \underbrace{\text{Freq}(v) \cdot \mathbb{E}[\ell_t | x_t = v]}_{R(v)} \quad (2)$$

103 We formally define the **Rock Score** as $R(v) = \bar{\ell}_v \cdot \text{Freq}(v)$, where $\bar{\ell}_v$ is the empirically estimated mean
 104 KL loss of token v at the final checkpoint. The Rock Score $R(v)$ is therefore not an approximation,

but the exact contribution of token type v to the total loss that OPD minimizes. Ranking tokens by $R(v)$ is the unique additive ranking that strictly agrees with the OPD objective.

The $\text{Freq}(v)$ factor acts as a natural regularizer: it up-weights tokens whose mean-loss estimate converges quickly under heavy sampling, while heavily discounting rare tokens whose mean is dominated by small-sample variance. Detection via $R(v)$ thus simultaneously aligns with the theoretical training objective while remaining robust to the empirical estimation noise inherent in natural language generation.

Operationally, $R(v)$ prioritizes tokens that exhibit both a substantial occurrence count and an above-baseline per-token loss. Rare tokens with extreme observed losses receive appropriately low scores because their $\text{Freq}(v)$ weight is minimal. As demonstrated empirically in Figure 2, the top-ranked tokens cluster in the mid-to-high frequency range and sit at the upper edge of the loss distribution within their respective frequency bands, successfully filtering out the rare-but-extreme statistical outliers.

2.3 Empirical Analysis

Rock tokens exhibit persistently high expected KL divergence between student π_θ and teacher π_T after on-policy distillation. We estimate this per-token loss $\hat{\ell}_v = \mathbb{E}[D_{\text{KL}}(\pi_\theta(\cdot | x_{<t}) \| \pi_T(\cdot | x_{<t})) | x_t = v]$ using student rollouts.

Setup The student (Qwen3-4B-Instruct) is on-policy distilled from the teacher (Qwen3-30B-A3B-Instruct) using OpenThoughts. Per-token KL is sampled from OpenThoughts and evaluated on $N=500$ MATH-500 problems (full details in Section 5).

Observations Figure 2 illustrates the distribution and density of the identified Rock Tokens within the student’s generated trajectories. Panel (a) displays the absolute count of Rock Tokens per sequence, while Panel (b) normalizes this count as a percentage of the total sequence length. Strikingly, the data reveals that Rock Tokens are highly concentrated, comprising a median of 52.5% of the total tokens in every generated output.

Take-away The Pervasiveness of Rock Tokens: Although Rock Tokens represent a tightly constrained subset of the overall vocabulary (e.g., $K = 100$), their footprint during generation is massive. Over half of the model’s reasoning trajectory consists of these stubborn structural and formatting tokens, demonstrating that the primary locus of student-teacher disagreement occurs consistently throughout the core scaffolding of the output.

2.4 Determining the Rock Token Cutoff

Defining the boundary of the Rock Token set $\mathcal{R}$ requires choosing a cutoff K that balances *selection stability* (reproducibility across samples) against *loss coverage* (fraction of optimization loss captured). A small K yields a stable set but low coverage, while a large K captures more loss but degrades under sampling noise.

Empirical sweeps (Figures 1b and 1c) identify an optimal intersection at $K = 100$. This boundary accounts for roughly 60% of the corpus-level KL divergence while maintaining high stability across subset sizes. We therefore adopt $K = 100$ to ensure a mathematically reproducible vocabulary for subsequent analyses. Comprehensive stability metrics and sweeping procedures are detailed in Appendix E.

2.5 Analysis of Rock Tokens

We decoded the top- $K=100$ Rock Tokens into their string forms using the Qwen3 tokenizer. Despite being identified by a purely numerical criterion, the resulting set is functionally coherent: Rock Tokens consistently mark structural or discourse boundaries in mathematical reasoning.

As detailed in Appendix B, four functional clusters dominate $\mathcal{R}$: (1) **LaTeX and math delimiters**, (2) **Markdown and whitespace structure**, (3) **Discourse markers** (e.g., “So”, “Wait”), and (4) **Digits**. In contrast, frequency-matched controls ($\mathcal{S}_{\text{ctrl}}$) predominantly consist of content-bearing tokens (e.g., nouns, verbs) used at non-boundary positions.

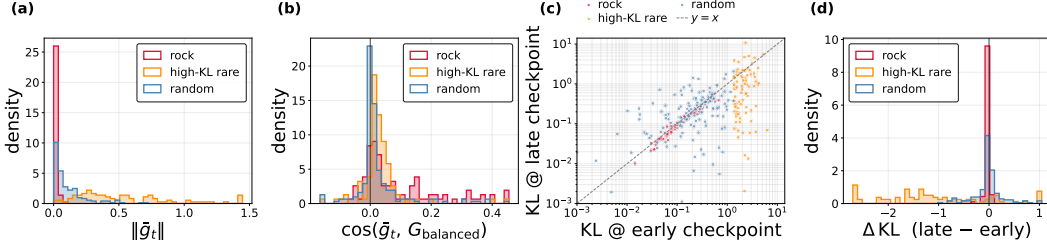


Figure 3: **Per-token gradient geometry and persistence under training.** (a) Per-token logit-gradient magnitude $\|\bar{g}_t\|$ by group: rocks are an order of magnitude smaller than rare high-KL tokens. (b) Cosine alignment with the frequency-balanced descent direction G_{balanced} : rocks are positively aligned, with a tail reaching $\cos > 0.3$. (c) Per-token mean KL paired across two training checkpoints (log-log). Points below the diagonal indicate KL reduction during further training; rocks lie on the diagonal, rare high-KL tokens fall below. (d) Distribution of $\Delta\text{KL} = \text{KL}_{\text{late}} - \text{KL}_{\text{early}}$ by group: rocks concentrate at $\Delta\text{KL} \approx 0$ (persistent); rare high-KL tokens have large negative ΔKL (learned).

Take-away Structure Over Content: The clustering of $\mathcal{R}$ at boundaries indicates student-teacher divergence centers on **how to structure** reasoning rather than **what content** to generate. Rock Tokens are a signature of the on-policy regime, marking where the student systematically resists the teacher’s discourse style.

149

150 2.6 Empirical Observations on Gradient Geometry and Persistence

151 Having established that Rock Tokens primarily comprise syntactic and structural scaffolding, we
 152 address **RQ1: Do Rock Tokens still provide useful learning signals during optimization?** To
 153 determine whether their persistent high loss effectively translates into steady directional guidance or
 154 merely stagnant optimization noise, we characterize their training dynamics and geometric properties
 155 against rare high-KL tokens and random tokens (illustrated in Figure 3).

156 To formally evaluate this, we analyze the geometry of per-token gradients. For each token type
 157 t with n_t occurrences, we compute the mean per-occurrence reverse-KL gradient in logit space,
 158 $\bar{g}_t = \frac{1}{n_t} \sum_i g_i$. We then decompose its contribution to the global descent as:

$$\text{contrib}(t) = n_t \cdot \|\bar{g}_t\| \cdot \cos(\bar{g}_t, G_{\text{balanced}}), \quad (3)$$

159 where $G_{\text{balanced}} = \sum_t \bar{g}_t$ is a frequency-balanced reference that grants every token type equal
 160 weight, deliberately removing the trivial frequency-weighted dominance of common tokens. This
 161 decomposition yields several key insights corresponding to our empirical observations:

162 **Low Gradient Magnitude vs. High Frequency (Panel a):** Rock tokens are characterized by
 163 substantially *smaller* per-occurrence gradient magnitudes than rare high-KL tokens (median $\|\bar{g}_t\| \approx$
 164 0.016 versus ≈ 0.54 ; Mann-Whitney $p < 10^{-30}$). However, the multiplicative decomposition in
 165 Eq. 3 shows that rock dominance arises from the *frequency* factor (n_t): because of their massive
 166 occurrence rate in the training corpus, this small per-token signal accumulates into a dominant
 167 aggregate force on the weight updates.

168 **Alignment with Global Optimization (Panel b):** Counter-intuitively, the gradient direction of Rock
 169 Tokens ($\bar{g}_t$) exhibits non-trivially positive directional alignment with the balanced global gradient
 170 G_{balanced} (median $\cos \approx 0.040$ versus ≈ 0.025 for high-KL and ≈ 0.006 for random tokens), with a
 171 sub-population reaching $\cos > 0.3$. This means that each rock contributes a small but consistently
 172 aligned signal at many positions, whereas rare high-KL tokens contribute large but direction-distinct
 173 signals at very few. From the teacher’s perspective, Rock Tokens do indeed point toward a “correct”
 174 optimization path for the overall objective.

175 **Persistence During Training (Panel c & d):** To confirm that this directional alignment translates
 176 into resistance to further learning, we compare per-token KL between an early and late checkpoint.
 177 Rare high-KL tokens experience a substantial reduction in KL (median $2.02 \rightarrow 0.85$, $\sim 58\%$;
 178 Wilcoxon $p < 10^{-8}$), moving significantly below the $y = x$ diagonal. In contrast, Rock Tokens
 179 remain essentially unchanged (median $0.21 \rightarrow 0.19$), clustering sharply along the diagonal with the

180 distribution of ΔKL concentrated exactly at zero (Panel d). Random control tokens show a symmetric,
181 noisy distribution with no net change.

182 This optimization stagnation is geometrically consistent with our gradient findings: rare high-KL
183 tokens possess direction-distinct gradients that focused updates can isolate and resolve. Conversely,
184 because rocks contribute highly aligned gradients, their targeted reduction would be indistinguishable
185 from following the global descent. Thus, they receive no separable, token-specific learning signal to
186 overcome their inherent loss.

187 While our gradient decomposition explains the geometry of this stagnation, the underlying mechanical
188 cause remains open. We hypothesize this paradox is driven by adaptive optimizers under heavy-tailed
189 class imbalance [9]. Unlike in pre-training—where Adam beneficially suppresses frequent tokens to
190 avoid overfitting structural noise—On-Policy Distillation often requires updating these exact high-
191 frequency structural tokens to match a teacher’s formatting or style. By inflating the second-moment
192 estimates for ubiquitous Rock Tokens, Adam systematically neutralizes their effective learning rate,
193 trapping the student in pre-existing habits despite possessing accurate directional gradients. We leave
194 the formal verification of this optimizer-suppression hypothesis to future work.

Take-away The Gradient Paradox: Rock Tokens *do* provide useful learning signals, generating
gradients that are well-aligned with the global optimization target (Panel b). However, they remain
stagnant and "untrained" throughout the OPD process (Panel d). This paradox suggests that their
persistence is not due to a lack of constructive signal, but rather the student model’s internal
optimization bias, which stubbornly prioritizes pre-existing structural scaffolding over the teacher’s
stylistic direction because the rock-token signal is mathematically inseparable from the global
descent direction.

196 3 Pillar Tokens: The Scaffolding of Fragile Reasoning

197 3.1 Existence as Necessity: The Functional Value of Rock Tokens

198 We have observed that **Rock Tokens** remain resistant to optimization throughout the entire OPD
199 process. To understand the origin of this resistance, we hypothesize that these tokens serve as the
200 structural “pillars” of the model’s reasoning process. Under this view, the student model may actively
201 protect these tokens from optimization to prevent its internal reasoning framework from collapsing,
202 even when they conflict with the teacher’s distribution. This leads to our **RQ2: Does Rock Token**
203 **persistence stem from student path dependency?**

204 3.2 Causal Probing via Inference-time Knockout

205 To move beyond distributional metrics and pinpoint the exact functional value of individual tokens,
206 we employ an inference-time **knockout** procedure. We probe the causal contribution of a token v by
207 observing how its absence affects the model’s accuracy on a benchmark $\mathcal{B}$. Specifically, we form a
208 *knockout policy* $\pi_{\theta}^{\setminus v}$ by forcing the logit of token v to $-\infty$ at every decoding step. We then measure
209 the **causal delta**:

$$\Delta_{\mathcal{B}}(v) = \text{Acc}_{\mathcal{B}}(\pi_{\theta}^{\setminus v}) - \text{Acc}_{\mathcal{B}}(\pi_{\theta}). \quad (4)$$

210 A token v is identified as a **Strong Pillar** if its removal causes a significant degradation ($\Delta_{\mathcal{B}}(v) \leq$
211 -0.01). We screen a pool of $|\mathcal{R}| = 200$ candidates, extending beyond the core $K = 100$ set to ensure
212 a wider search for these rare causal effects (see Appendix D.4 for statistical and operational details).

213 3.3 Are Rock Tokens Pillars? Findings and Orthogonality

214 The distribution of $\Delta_{\mathcal{B}}(v)$ in Figure 4 reveals the underlying nature of the Rock Token set. Our
215 investigation yields three primary observations:

216 **Pillars are rare but vital:** Only 3.5% of Rock Tokens on MATH-500 and 1.5% on IFEval qualify
217 as Strong Pillars. While most high-KL tokens are neutral residuals, a small subset is absolutely
218 indispensable for the model’s reasoning trajectory. **Directional Asymmetry:** The complete absence
219 of Strong Stumbling Blocks suggests that the student’s “stubborn” deviations from the teacher are
220 rarely detrimental to its own logic; they are either necessary anchors or harmless stylistic preferences.

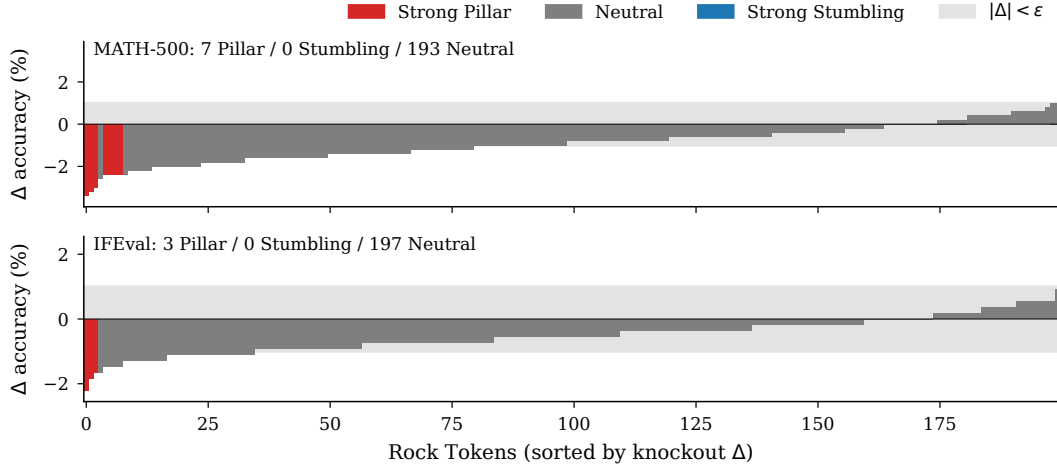


Figure 4: **Knockout effect on the $|\tilde{\mathcal{R}}| = 200$ screened Rock-Token candidates, sorted by Δ .** Each bar is a single candidate; height is the accuracy change when its logit is masked at decode time. The shaded grey band marks the categorization threshold $|\Delta| < \epsilon = 0.01$. Bars outside the band that pass the paired-bootstrap test ($\alpha = 0.05$, 10,000 resamples) are coloured by category (Strong Pillar in red; Strong Stumbling in blue). The vast majority of candidates are Neutral on both benchmarks; Pillars occupy a short tail and the symmetric Strong-Stumbling side is empty — the visual evidence underlying the sign-asymmetry test formalized below.

221 **Orthogonality to Traditional Metrics:** To examine if Pillarhood can be predicted by standard
 222 signals, we analyzed the correlation between $\Delta_{\mathcal{B}}(v)$ and several distributional metrics, including
 223 student/teacher entropy, log-frequency, and residual KL. Crucially, none of these predictors exhibit
 224 a significant correlation with a token’s causal importance ($|r| < 0.07$, see Appendix D.5 for full
 225 analysis and scatter plots). This demonstrates that Pillarhood is a property orthogonal to existing
 226 importance taxonomies [19], suggesting that importance schemes keyed to loss or entropy signals
 227 risk suppressing the very tokens the student model cannot reason without.

228 **Take-away The Concentration of Necessity.** While Rock Tokens as a population are indispensable for training stability, their causal necessity at inference is highly concentrated: only a tiny fraction (1.5%–3.5%) act as true **Pillars**. This refutes the hypothesis that the student relies on the *entire* high-loss set to maintain reasoning; instead, the model appears to protect a wide "buffer zone" of persistent residuals to ensure a few critical, semantic anchors remain intact.

229 4 The genuine contribution of Rock Tokens

231 4.1 RQ3: What is the genuine functional 232 contribution of Rock Tokens to model training?

233 The functional redundancy of Rock Tokens at inference prompts a critical question: do their persistent high-loss signals provide essential constraints, or are they merely **optimization noise**? We consider two competing hypotheses: (1) **Late-stage Saturation**, where these tokens act as early alignment drivers but become unproductive "Stumbling Blocks" as the model reaches structural limits; or
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 242 (2) **Inherent Inutility**, where they represent superficial stylistic gaps with no reasoning value from the outset.

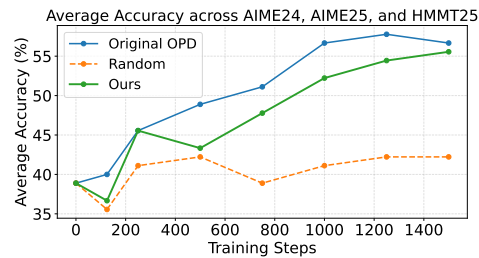


Figure 5: Average accuracy across AIME24, AIME25, and HMMT25 during OPD training. Each 200 training steps correspond to 8,000 prompts, with 4 rollouts per prompt.

243 To adjudicate between these views, we employ a **selective distillation strategy** by freezing the
 244 gradients of Rock Tokens. By comparing training trajectories with and without these signals, we aim
 245 to determine whether these persistent residuals are indispensable functional anchors or a detrimental
 246 optimization burden that consumes gradient bandwidth without yielding performance gains.

247 4.2 Selective Distillation via Token-Level Reweighting

248 To evaluate the functional necessity of these persistent residuals, we introduce a token-level reweight-
 249 ing mechanism to the OPD objective:

$$\mathcal{L}_{\text{weighted}} = \mathbb{E}_{x \sim \pi_\theta} \sum_{t=1}^T w(x_t) \cdot \ell_t, \quad \text{where } w(v) = \begin{cases} \lambda, & v \in \mathcal{R} \\ 1, & v \notin \mathcal{R} \end{cases} \quad (5)$$

250 Here, $\mathcal{R}$ is the Persistence Set defined in Section 2.2, and $\lambda \in [0, 1]$ serves as a scaling factor to
 251 modulate optimization pressure. This formulation allows us to test whether the gradient bandwidth
 252 consumed by tokens with high Rock Scores $R(v)$ is productive for learning or merely an optimization
 253 burden.

254 Our core experimental design employs three regimes by manipulating λ . The **Baseline OPD**
 255 ($\lambda = 1$) follows standard distillation without intervention. **Rock-Freeze (Our Method)**: Based
 256 on the identified set $\mathcal{R}$, we adopt a radical "Just Not Train" approach by setting $\lambda = 0$. This
 257 strategy effectively **masks the gradients** of all $v \in \mathcal{R}$, allowing us to determine if these tokens
 258 are indispensable functional anchors or simply unproductive residuals that can be bypassed to save
 259 computational budget. Finally, the **Freq-Matched Freeze (Random)** applies the same $\lambda = 0$ strategy
 260 to a randomly sampled set $\mathcal{S}$ that strictly matches the **token frequency distribution** of $\mathcal{R}$. By
 261 comparing these three settings, we can decouple the impact of structural Rock Tokens from stochastic
 262 noise and frequency-related artifacts.

263 4.3 Impact of Rock Tokens on Reasoning Performance

264 Figure 5 illustrates the performance trajectories across AIME24, AIME25, and HMMT25 under dif-
 265 ferent intervention regimes. Our analysis reveals three key findings. First, while Rock Tokens exhibit
 266 persistent high loss, they are not merely "noise traps"; their inclusion provides essential gradients
 267 that anchor the optimization process, as evidenced by the performance gap between the *Original*
 268 *OPD* and *Ours*. Second, while reducing total gradient signals is generally detrimental—shown by the
 269 severe degradation in the *Random* baseline—selectively down-weighting Rock Tokens is significantly
 270 safer, maintaining a much higher accuracy ceiling. Third, in our setup, Rock Tokens account for
 271 approximately 50% of the total output tokens; by mitigating the optimization pressure on these
 272 residuals, our empirical results demonstrate a $1.7\times$ increase in overall training wall-clock speed
 273 without substantial loss in reasoning integrity.

Take-away The Efficiency-Performance Frontier: While Rock Tokens provide necessary
 structural gradients, they exhibit significantly diminishing marginal utility. Our results demonstrate
 that applying a freeze-weighting to these 30% high-cost tokens yields a $1.7\times$ **wall-clock speedup**
 with minimal impact on reasoning integrity. This confirms that while Rock Tokens are not entirely
 disposable, **non-uniform gradient allocation** can effectively bypass computational redundancy in
 on-policy alignment.

275 5 Experiment Setups

276 5.1 Off-policy and On-policy Distillation

277 We implement a two-stage distillation pipeline via the **KDFlow** framework [22] to transfer knowledge
 278 from a teacher, **Qwen3-30B-A3B-Instruct-2507** (MoE, 3B active), to a student, **Qwen3-4B-Instruct-**
 279 **2507** [20], with thinking mode disabled for both. **Stage 1 (Off-Policy)**: To achieve initial *all-*
 280 *vocabulary alignment*, the student is trained for one epoch on 20k teacher-generated solutions from
 281 **OpenThoughts3** [6] using a forward KL objective ($\text{kd_ratio} = 0.5$) with cross-entropy. **Stage 2**
 282 **(On-Policy)**: Starting from the Stage 1 checkpoint, the student samples 4 rollouts each prompt from
 283 a separate 10k prompts as a checkpoint, (7 checkpoints in total) and is trained for one epoch on its

own trajectories using pure reverse KL ($kd_ratio = 1.0$). Unless specified, all other hyperparameters follow KDFlow defaults; see Appendix C for full implementation details.

5.2 Evaluation

We employ the LM-EVALUATION-HARNESS framework [5] for standardized assessment across all benchmarks in a zero-shot setting. To ensure statistical robustness, we report the **Pass@1** accuracy averaged over five independent runs, accounting for variance in decoding.

Our evaluation suite focuses on challenging competitive mathematics, comprising **AIME 24**, **AIME 25**, and **HMMT 25-Feb** from MathArena [3]. Each benchmark contains 30 problems; we report the aggregated average score across these 90 tasks as the primary performance metric in our study. We extend our analysis to MATH-500 [11] and IFEval [24] to obtain a robust sample size for token-level statistical evaluation.

6 Related Works

6.1 On-Policy Distillation for LLM Post-Training

On-policy distillation evolves beyond SFT and RLVR by combining trajectory exploration with dense, token-level guidance [12]. Recent large-scale systems like DeepSeek-V4 [2], MiMo [18], and Qwen-3 [20] demonstrate its efficacy as a post-training stage, while self-distillation studies [23, 14, 8] suggest OPD is a general mechanism for amplifying latent model capabilities rather than mere imitation. However, the success of OPD is often hindered by teacher-student misalignment. While prior research examines this incompatibility through global factors like vocabulary, reasoning style, or model family [4, 10], our work focuses on a fine-grained source of stagnation. We investigate **Rock Tokens**—persistent high-loss positions that resist alignment even after training plateaus—to determine when OPD supervision provides productive signals versus unproductive noise.

6.2 Token-Level Learning Dynamics in Reasoning Models

Recent studies demonstrate that reasoning in LLMs is disproportionately supported by critical “anchor” tokens, the perturbation of which during inference significantly degrades model performance [7]. In parallel, RLVR research reveals that training is non-uniform [21, 1, 17], concentrating optimization on high-entropy “forking tokens” that act as decisive branching points for reasoning gains [15], and further analysis confirms the overall token distribution drift mainly happens on some sparse but crucial tokens [13]. Although OPD’s dense supervision suggests high-loss tokens drive training, their specific contribution mechanism remains unclear—especially when they persist as **Rock Tokens**. It is unknown whether these recalcitrant positions are indispensable reasoning pillars or merely unproductive residues. We address this gap by systematically analyzing their gradient dynamics and training effects to determine their true role in the OPD process.

7 Conclusion

This paper investigates the role of high-loss tokens in On-Policy Distillation, revealing that while they occupy a dominant position in the loss landscape, they are not the decisive factor for performance gains. Our analysis identifies a distinct category of persistent tokens, termed **Rock Tokens**, despite their significant gradient magnitude and apparent “effort” during optimization, Rock Tokens fail to provide constructive guidance for the model’s reasoning capabilities. Notably, our intervention experiments demonstrate that applying a freeze-weighting to these tokens—thereby deprioritizing them during the optimization update—results in negligible impact on downstream performance while achieving a $1.7\times$ increase in training wall-clock speed. These findings challenge the conventional wisdom that aligning every high-loss token is indispensable for learning. Instead, we advocate for a more resource-efficient, token-selective approach that bypasses the computational redundancy of low-utility tokens to optimize the efficiency-performance frontier in on-policy alignment.

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400 A Limitations

401 First, our empirical evaluations rely predominantly on competitive mathematical reasoning; the
 402 functional role of Rock Tokens may differ in open-ended generation or coding tasks where structural
 403 boundaries are less rigid. Second, our experimental setup utilizes a specific 30B-to-4B distillation
 404 pairing, leaving the persistence and geometry of these tokens across different architectural families
 405 or vastly larger scales an open question. Finally, to isolate the functional impact of Stumbling
 406 Blocks, our mitigation strategy employs a strict binary gradient-freeze mask. While effective for
 407 demonstrating causal necessity, this approach is a blunt instrument; future work should explore
 408 dynamic reweighting schemes or soft-penalties that gracefully adapt to token-level loss trajectories
 409 throughout the optimization process.

410 A.1 Ethics Discussion and Risk Assessment

411 This work studies model distillation and training dynamics, and does not introduce a deployed
 412 system, human-subject data, or sensitive personal data. Its direct societal risks are therefore limited.
 413 However, because improvements in language model training may be used in both beneficial and
 414 harmful downstream applications, we acknowledge potential dual-use risks, including misuse for
 415 generating low-quality or misleading content.

416 We mitigate these risks by limiting the paper to research analysis and experimental evaluation.
 417 Any released code, models, or processed artifacts will be accompanied by appropriate documenta-
 418 tion, intended-use information, and license details. All pretrained models, datasets, and software
 419 frameworks used in this work are credited, and their licenses and terms of use are respected.

420 B Analysis on Rock Tokens

421 We decode the top- $K=100$ rock tokens and their frequency-matched controls $\mathcal{S}_{\text{ctrl}}$ into their string
 422 forms using the public Qwen3 tokenizer. Despite being identified by a purely numerical criterion
 423 ($R(v)$), the resulting set is functionally coherent: rock tokens fall into a small number of identifiable
 424 categories, all of which mark structural or discourse boundaries in mathematical reasoning output.

425 Four functional clusters dominate $\mathcal{R}$.

- 426 1. **LaTeX and math delimiters** — `$`, `\\`, `$$`, `=`, `^`, `{`, `}`, `frac`, `(`, `)`. These mark transitions
427 into and out of math mode and operate at the syntactic boundary between prose and formula.
- 428 2. **Markdown and whitespace structure** — `**`, `\n`, `\n\n`, `:\n\n`, `—\n\n`, `###`. These delimit
429 logical units (paragraphs, section headers, bold spans).
- 430 3. **Discourse markers at sentence-initial position** — `So`, `Let`, `We`, `But`, `Now`, `Wait`,
431 `Then`, `Since`, `This`. These open new reasoning steps and signal turns in the chain of
432 thought.
- 433 4. **Digits** — every digit 0 through 9 appears in $\mathcal{R}$, each with frequency in the thousands and a
434 non-trivial mean per-token KL. These are tokens that punctuate quantitative statements.

435 Contrast with the frequency-matched control set. The control set $\mathcal{S}_{\text{ctrl}}$, sampled at the same frequency
436 distribution, is qualitatively different: it consists predominantly of content-bearing tokens used at non-
437 boundary positions — common nouns and verbs of mathematical discourse (e.g. `triangle`, `equation`,
438 `region`, `expression`, `must`, `find`, `check`), prepositions and conjunctions used mid-clause (`as`,
439 `between`, `first`, `right`), and single-letter variable tokens (`a`, `b`, `i`, `f`, `y`). The frequency-matching
440 procedure thus isolates the relevant axis: rocks and controls share the quantity of supervision they
441 receive during OPD, but differ sharply in their syntactic role.

442 **Interpretation** The clustering of $\mathcal{R}$ at structural boundaries indicates that the residual student–teacher
443 divergence after on-policy distillation is concentrated at the points where the model decides *how to*
444 *structure* its reasoning output, rather than *what content* to emit. Decisions like `open math mode`,
445 `begin a new paragraph`, `insert a section header`, or start a new reasoning step with `So` versus
446 `Then`, are precisely the positions where the OPD loss persists, even after extensive distillation. This
447 observation suggests that rock tokens are not arbitrary statistical outliers but a consistent signature
448 of the on-policy distillation regime: they identify the structural/discourse decisions that the student
449 systematically fails to align with the teacher.

450 C Training and Evaluation Details

451 C.1 off policy and on policy training

452 **Training Details.** We distill from Qwen3-30B-A3B-Instruct-2507, a Mixture-of-Experts teacher
453 with $\sim 30\text{B}$ total and $\sim 3\text{B}$ active parameters, into Qwen3-4B-Instruct-2507 as the student, using the
454 KDFlow framework. Both models are run with thinking mode disabled (`enable_thinking=False`).
455 Training proceeds in two stages on a single node of $4 \times \text{H100}$ (80 GB) GPUs with FSDP2, bf16, and
456 gradient checkpointing.

457 **Stage 1 – Off-policy KD.** We first sample 20k teacher responses on math prompts drawn from
458 OpenThoughts3 (temperature = 0.6, top_p = 0.95, max_new_tokens = 16384, TP = 2 on $2 \times$
459 H100). The student is then trained on these (prompt, teacher-response) pairs with a per-token KL
460 distillation loss combined with cross-entropy at kd_ratio = 0.5 (vanilla KD, forward KL). We use
461 AdamW with learning rate 2×10^{-5} , 5% linear warmup, global batch size 128 (micro-batch 1), max
462 sequence length 16384, sample packing, and ring attention of size 2, for 1 epoch.

463 **Stage 2 – On-policy KD.** Initialized from the Stage-1 checkpoint, the student generates 4 rollouts
464 per prompt (temperature = 1.0, top_p = 1.0, generate_max_len = 8000, prompt_max_len = 800)
465 on a separate 10k-prompt slice (positions 20k–30k of the same source), and is distilled toward the
466 teacher’s token distributions on those rollouts. We use vanilla KD with reverse KL at kd_ratio = 1.0
467 (pure distillation, no CE), learning rate 2×10^{-6} , 5% linear warmup, gradient clipping at 1.0, global
468 batch size 4 (micro-batch 1), and 1 epoch. The rollout engine uses 1 engine with TP = 2 and the
469 teacher uses TP = 4; both engines share GPUs with the trainer via offload-to-CPU sleep/wakeup
470 (`teacher_enable_sleep=True`, `rollout_enable_sleep=True`).

471 **Defaults.** All other hyperparameters are KDFlow defaults; we did not tune optimizer betas, weight
472 decay, dropout, or KD temperature (kept at 1.0).

D Pillar Correlation and Multiple-Testing Analysis

D.1 Pillar Correlation Analysis

Table 1 reports the full set of Pearson correlations between the inference-time knockout effect $\Delta_{\mathcal{B}}(v)$ (Eq. 6) and nine candidate token-level predictors, measured over the $|\tilde{\mathcal{R}}| = 200$ screened Rock-Token candidates of §D.4 and reported separately for MATH-500 and IFEval. The figure in the main text (Figure 6) visualizes a representative subset of six of these predictors on MATH-500; this table is the complete record, including IFEval and the three loss-based features (pre-/post-OPD KL and KL improvement). Across all eighteen (*predictor, benchmark*) pairs, no correlation reaches $|r| > 0.075$ and no p -value falls below 0.30. The largest observed magnitude on either benchmark is the IFEval correlation with KL improvement ($r = +0.071, p = 0.32$), which is well within the noise band expected at $n = 200$. We therefore cannot reject the null hypothesis $r = 0$ for any predictor on either benchmark, supporting the claim in §D.5 that Pillarhood is statistically orthogonal to the entropy-, frequency-, and divergence-based importance signals used by prior work [15, 19, 21, 17].

Table 1: Pearson correlation between the per-token knockout effect $\Delta_{\mathcal{B}}(v)$ and nine candidate predictors over the $|\tilde{\mathcal{R}}| = 200$ screened Rock-Token candidates, reported for MATH-500 and IFEval. p -values are two-sided. No correlation reaches $|r| > 0.075$ on either benchmark; no p -value falls below 0.30.

Predictor	MATH-500		IFEval	
	r	p	r	p
Frequency	+0.015	0.832	+0.030	0.677
Rock count	+0.013	0.855	+0.023	0.751
Rock rate	−0.022	0.760	−0.047	0.512
Pre-OPD KL	−0.017	0.808	−0.015	0.833
Post-OPD KL	+0.010	0.883	−0.048	0.503
KL improvement	−0.062	0.386	+0.071	0.320
Teacher entropy	+0.063	0.378	−0.036	0.611
Pre-OPD student entropy	+0.059	0.406	−0.069	0.331
Post-OPD student entropy	+0.056	0.432	−0.037	0.604

D.2 Multiple-Testing Analysis for the Per-Token Pillar Criterion

The per-token Pillar criterion of §D.3 applies a paired bootstrap test at $\alpha = 0.05$ independently to each of the $|\tilde{\mathcal{R}}| = 200$ screened Rock-Token candidates, conjoined with the effect-size filter $|\Delta_{\mathcal{B}}(v)| \geq \varepsilon$ ($\varepsilon = 0.01$). Because ε is approximately at the per-token sampling-error scale, the conjunction does not appreciably reduce the per-test Type-I rate, and $200 \times 0.05 = 10$ rejections are expected under the global null. We report below how the per-token identifications behave under standard family-wise and false-discovery corrections, and the global sign-asymmetry test on which the population-level claim of §D.4 rests.

Per-token corrections. Bonferroni control of the family-wise error at $\alpha = 0.05$ requires $p < \alpha/200 = 2.5 \times 10^{-4}$. The smallest observed p -values are 0.0038 (MATH-500, “certain”) and 0.0016 (IFEval, “-like”); neither passes Bonferroni, and consequently no token does. Benjamini–Hochberg control of the false discovery rate at $q \in \{0.05, 0.10, 0.20\}$ likewise produces zero rejections on either benchmark: the BH cutoff $p_{(k)} \leq (k/n)q$ is not crossed even for the smallest- p token, because $1/200 \times 0.20 = 10^{-3} < 0.0016$. The per-token Pillar identifications are therefore *exploratory candidates*, not confirmatory rejections in the multiple-testing sense.

Global sign-asymmetry test. The exploratory candidates do, however, exhibit a non-trivial structural property: the sign of the significant rejection is consistent across benchmarks. Of the seven MATH-500 candidates, all seven satisfy $\Delta_{\text{MATH500}}(v) < 0$ (Pillar direction); of the three IFEval candidates, all three satisfy $\Delta_{\text{IFEval}}(v) < 0$. Combined, the rejection sign-distribution is 10 negative, 0 positive. Under the global null in which every per-token rejection is a chance event, the rejection sign should be Bernoulli($1/2$), and the probability of observing all 10 on one side is $2^{-10} \approx 9.8 \times 10^{-4}$ (two-sided exact binomial $p = 1.95 \times 10^{-3}$). On MATH-500 alone the analogous sign test is 7/0

with two-sided $p = 0.016$; on IFEval alone 3/0 with $p = 0.25$. The combined test is a single global test that does not inflate with the number of tokens screened, and it is the result on which §D.4 relies for the population-level claim that a non-empty subset of Rock Tokens is causally necessary at decode time.

Summary. The reader is advised to read the per-token Pillar identifications of §D.4 (e.g. “certain”, “strategic”, “Initialize”) as the strongest individual candidates from a population-level effect, rather than as 7 independently-confirmed discoveries. The empirical contribution of this section that survives strict multiple-testing scrutiny is the sign asymmetry, which is incompatible with the multiplicity-noise null at $p \approx 0.002$.

Stable-core robustness check. The screening pool of §D.3 is the top- $K_{\text{screen}} = 200$ count-ranked extension of $\mathcal{R}$, while the definitional Rock-Token set fixed by the stability/coverage analysis of §2 is the top- $K = 100$. The 10 Strong Pillars from the bootstrap distribute as follows over the count-rank index. On MATH-500 (ranks reported on the top-200 list): “certain” (16), “Initialize” (93), “strategic” (99), “Do” (109), “programs” (143), “tech” (166), “arms” (185). On IFEval: “-like” (35), “tools” (64), “trade” (69). Six of the ten lie within the $K = 100$ core, including the smallest- p token on each benchmark (“certain” at $p = 0.0038$ on MATH-500 and “-like” at $p = 0.0016$ on IFEval). The remaining four lie in the rank-109-to-185 regime, which §2 flags as below the Jaccard-stability sweet spot; we therefore caution that those four are at greater risk of being corpus-specific. The sign-asymmetry test of the previous paragraph is robust to this concern: restricted to the $K = 100$ stable core, it remains 6/0 negative-versus-positive, with two-sided exact-binomial $p = 2^{-6} \times 2 = 3.1 \times 10^{-2}$.

D.3 Definition and Identification of ‘Pillar Tokens’

To investigate this, we operationally categorize Rock Tokens based on their causal impact at inference time. **Pillar Tokens** are defined as those whose removal or perturbation degrades task accuracy, suggesting the model *cannot reason without them*, while the remainder are deemed functionally neutral.

Intuition. The core idea is that if a token is truly load-bearing for reasoning, deleting it from the decoder should break the trajectory; if it only *looks* “hard” due to high loss but lacks functional utility, its deletion should change nothing. We operationalize Pillarhood as exactly this causal, inference-time property. Given the post-OPD student π_θ and a candidate $v \in \tilde{\mathcal{R}}$ (screening pool defined below), we form a knockout policy $\pi_\theta^{\setminus v}$ that sets the logit of v to $-\infty$ at every decoding step and record the accuracy delta on a held-out benchmark $\mathcal{B}$:

$$\Delta_{\mathcal{B}}(v) = \text{Acc}_{\mathcal{B}}(\pi_\theta^{\setminus v}) - \text{Acc}_{\mathcal{B}}(\pi_\theta). \quad (6)$$

A candidate v is a **Strong Pillar** on $\mathcal{B}$ iff (i) $\Delta_{\mathcal{B}}(v) \leq -\varepsilon$ ($\varepsilon = 0.01$) and (ii) a paired bootstrap on per-problem indicators (10,000 resamples, $\alpha = 0.05$) rejects $H_0: \Delta = 0$; symmetric thresholds at $+\varepsilon$ define Strong Stumbling Blocks, and all other candidates are Neutral. Accuracies are estimated by sampling at $T=1.0$ and averaging over independent rollouts to keep single-trajectory variance below the $\sim 2\%$ effect scale.

The screening pool is a strict superset $\tilde{\mathcal{R}} \supset \mathcal{R}$ of size $K_{\text{screen}} = 200$ rather than the definitional $K = 100$ set $\mathcal{R}$ of §2.2: $K = 100$ is the stability/coverage sweet spot when $\mathcal{R}$ is treated as a vocabulary (qualitative analysis, gradient-zeroing in §4), but Pillarhood is a per-token causal property and the bootstrap is a search for individual rare-but-real effects, so we trade selection stability below rank 100 for a wider candidate pool. As a robustness check (Appendix D.2), 6 of the 10 Strong Pillars — including the strongest on each benchmark, “certain” on MATH-500 and “-like” on IFEval — lie within the $K = 100$ core. We probe the post-OPD student on two complementary benchmarks: MATH-500 [11] for the trained reasoning capability and IFEval [24] for instruction-following as an out-of-distribution control.

D.4 Are Rock Tokens pillar? Experiments and findings

Table 2 reports the full Pillar census from the bootstrap procedure across the $|\tilde{\mathcal{R}}| = 200$ screened candidates. The dominant feature of the per-token $\Delta_{\mathcal{B}}(v)$ distribution is not the tail but the centre:

Table 2: Pillar census over the $|\tilde{\mathcal{R}}| = 200$ screened Rock-Token candidates, by benchmark.

Benchmark	Strong Pillar	Neutral	Strong Stumbling	Baseline acc.
MATH-500	7 (3.5%)	193 (96.5%)	0 (0.0%)	0.750
IFEval	3 (1.5%)	197 (98.5%)	0 (0.0%)	0.752
Cross-task agreement: 190 N N, 7 P N, 3 N P, 0 P P				

on both benchmarks the great majority of candidates fall inside the ε -band, indicating no detectable causal effect of their removal (visualization in Fig. 4, Appendix D.2).

Multiple-testing considerations. At $|\tilde{\mathcal{R}}| = 200$ candidates, ~ 10 false rejections are expected by chance at $\alpha = 0.05$, and neither Bonferroni nor Benjamini–Hochberg ($q \leq 0.20$) retains any individual token (Appendix D.2). We therefore read the per-token identifications above as exploratory *candidate* Pillars. The population-level claim is carried by the *sign* of the rejections: all 10 fall on the Pillar (negative- Δ) side and zero on the Stumbling side, a 10:0 split with exact probability $2^{-10} \approx 0.002$ under the multiplicity-noise null — a single global test that does not inflate with the number of candidates screened.

Three observations follow. **First**, Pillars are rare: 3.5% on MATH-500 and 1.5% on IFEval — the persistent high-KL criterion of §2 over-selects causally relevant tokens by more than an order of magnitude. **Second**, Pillarhood is task-specific: no token is a Strong Pillar on both benchmarks ($P|P = 0$ in Table 2), so the role of a token type depends on the task. **Third**, Strong Pillars are content-bearing tokens at decision points, not the structural delimiters that dominate the surface clustering of $\mathcal{R}$ in §2: MATH-500 examples include “certain”, “strategic”, “Initialize”, “programs”, “Do”; IFEval includes “tools”, “trade”, “-like”. The symmetric Strong-Stumbling direction is empty under inference-time knockout, an asymmetry we revisit at training time in §4.

D.5 Does Rock Token follow previous findings on ‘Forking Tokens’?

Intuition. The dominant prior framing casts “important tokens” as high-entropy decision points where the policy splits between alternative reasoning continuations: forking tokens in RLVR [15], and the (entropy $\cup$ student–teacher-divergence) taxonomy formalized for OPD by TIP [19] and related token-reweighting work [21, 17]. If our Pillars were the inference-time projection of any of these training-time importance criteria, knockout $\Delta_{\mathcal{B}}(v)$ should be predicted by entropy, by KL, or by some combination thereof.

Findings. Figure 6 scatters MATH-500 knockout $\Delta_{\text{MATH500}}(v)$ against six candidate predictors — pre/post-OPD student entropy, teacher entropy, $\log_{10}$ frequency, rock rate, and post-OPD KL — and *none* reaches $|r| > 0.07$; the same conclusion holds on IFEval and on three additional loss-based predictors (Table 1, Appendix D.1). Pillars are scattered across the entire entropy range, not concentrated where forking tokens reside, and they are not the tokens with the largest residual KL — the criterion that $\mathcal{R}$ itself maximizes. Pillarhood is therefore an *orthogonal* cut: a causal, behaviour-defined property that the existing entropy- and divergence-based taxonomies fail to surface, with the immediate methodological corollary that any token-importance scheme keyed to those signals risks suppressing exactly the tokens the student cannot reason without.

Take-away Pillar Tokens are not Forking Tokens. Their causal role at inference is uncorrelated with student entropy, teacher entropy, frequency, or post-OPD KL ($|r| < 0.07$ on MATH-500), placing them outside the importance taxonomies of prior work [15, 19].

E Selection of Ratio

Choosing the selection size. The score $R(v)$ provides a ranking; we still need a cutoff K that determines $|\mathcal{R}|$. Two competing criteria bear on this choice. First, K should be small enough that the top- K set is stable — i.e., reproducible across resamplings of the rollout corpus. Second, K should be large enough that $\mathcal{R}$ accounts for a substantial fraction of the corpus-level KL, so that reweighting decisions made on $\mathcal{R}$ have meaningful effect on $\mathcal{L}_{\text{OPD}}$.

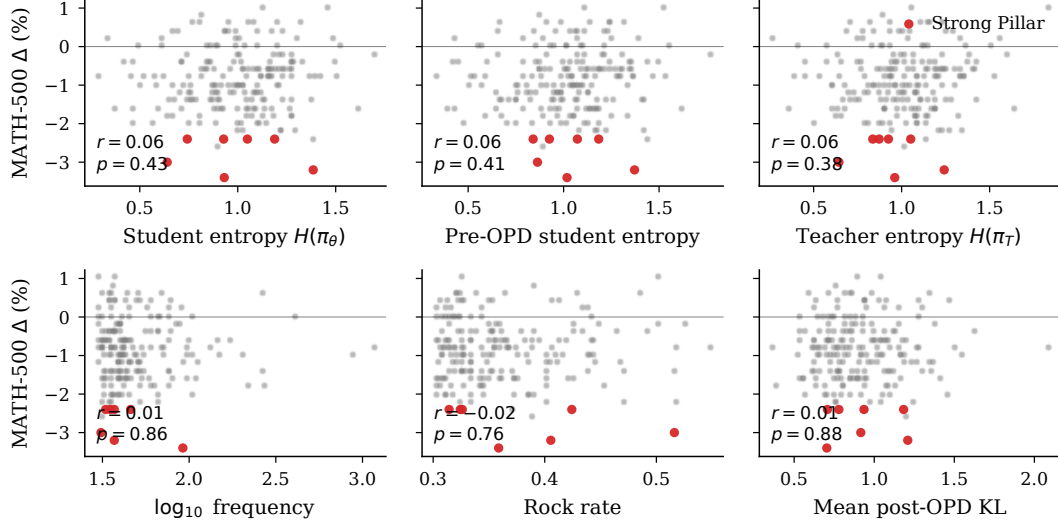


Figure 6: **Pillarhood is not predicted by entropy, frequency, or loss.** MATH-500 knockout Δ for each of the $|\tilde{\mathcal{R}}| = 200$ screened candidates (Strong Pillars in red) plotted against six candidate predictors: post- and pre-OPD student entropy, teacher entropy, log-frequency, rock rate, and mean post-OPD KL. Annotated r, p are Pearson correlations over all 200 candidates. None reach $|r| > 0.07$.

597 **Procedure** We sweep K jointly with sample size. For each $n \in 50, 100, 200, 300, 400$, we compute
 598 $\hat{R}(v)$ from the corresponding sub-corpus and select its top- K tokens; we then measure the Jaccard
 599 index of this set against the top- K obtained at $n = 500$. We additionally compute the cumulative
 600 coverage $\sum_{v \in \mathcal{R}} \hat{R}(v) / \mathcal{L}_{\text{OPD}}^{\text{corpus}}$ at $n = 500$ as K grows.

601 **Results** Figure 2(b) plots Jaccard against K , with one curve per n and an averaged curve. Average
 602 Jaccard exceeds 0.70 for $K \leq 100$ and decays toward a ≈ 0.57 floor for $K \geq 150$ — beyond which
 603 the additional tokens are no longer reliably reproduced across sub-corpora. Figure 2(c) overlays the
 604 corresponding coverage curve: at $K = 100$, $\mathcal{R}$ accounts for $\approx 60\%$ of the corpus-level KL while
 605 remaining within the stable regime. We therefore set $K = 100$ ($\approx 2.5\%$ of the observed vocabulary)
 606 as the *definitional* Rock-Token set $\mathcal{R}$ used wherever a stable, reproducible vocabulary is required —
 607 the qualitative cluster analysis below, and the training-time interventions of Section 4. This choice
 608 represents an explicit tradeoff: $K = 50$ would yield modestly higher stability ($J \approx 0.81$) but reduce
 609 coverage to $\approx 49\%$. For the per-token causal analysis of Appendix D.4 we instead *screen* a strict
 610 superset of size $K_{\text{screen}} = 200$: that section’s bootstrap is a search for individual rare-but-real Pillars,
 611 and the broader candidate pool both increases the chance of detection and lets us probe the regime
 612 in which selection stability has decayed below $J \approx 0.6$. The two cutoffs serve different purposes
 613 — the smaller K defines $\mathcal{R}$, the larger one widens the screening window — and we are explicit in
 614 Appendix D.3 about which is in force.

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